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Detection of Retinal Diseases in Retinal Optical Coherence Tomography Images

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***Abstract*— In biomedicine, disease image classification is a laborious, manual process. However, recent developments in computer vision have made applying artificial intelligence for the detection and categorization of medical images a more practical option. It is essential to analyse retinal illnesses utilising Optical Coherence Tomography (OCT) scans. For patients to survive, these disorders must be accurately detected and classified. The analysis of retinal illnesses is currently done by doctors on a regular basis by looking at the OCT pictures. The manual diagnosing process is laborious, though. In order to aid clinicians in their diagnosis, an automatic detection and classification technique for retinal illnesses has been developed in this work. Utilizing OCT scans, a deep multilayered Artificial Neural Network (ANN) has been employed to identify and categorise the retinal anomalies. On a 150 image open source retinal OCT dataset, the suggested technique has been used.**

Keywords*— Optical Coherence Tomography, Macular Hole, Pigment Epithelial Detachment, Central Serous Retinopathy, Artificial Neural Networks

I. INTRODUCTION

The recognition and diagnosis of anomalies in a variety of imaging modalities is greatly aided by classifiers in medical image processing. The demand for this classification has increased as automation is currently increasing in every sector of society. This work focus on the optical coherence tomography imaging modality for ophthalmic imaging, one of several imaging modalities for earlier diagnosis in ophthalmology. OCT uses an effective and cutting-edge method [1] that uses ultrasonic waves as both a source and a receiver to produce images of the retina's different layers [2]. Additionally, the fundus camera is a very effective technique for retinal imaging. But in order to grasp the real study of the fluid-based abnormality better, OCT is performed anytime a three-dimensional reconstructible image of the retina is needed.

A change in the fluid pattern of the different retinal layers is one of the early stages of blindness. Macular Hole (MH), one of these anomalies where the macular layer loses fluid, is one example. Two conditions that cause fluid to accumulate behind the retina in the eye are central serous retinopathy (CSR) and pigment epithelial detachment (PED), in which new blood vessels or tears grow and the RPE layer's fluid level rises.

The division of liquid filled layers, optical circles, and liquid filled layers is far more basic than the division of intraretinal layers, which was the focus of previous efforts in the investigation of OCT images [3] through [12]. The system is distinct in that it seeks to categorise the input photos as normal or abnormal based on a number of factors. Using criteria like sensitivity and specificity, a classifier's performance, like the SVM Classifier, will be assessed. From these parameters, the system's accuracy can be measured and quantified.

Classifiers in medical image processing play a important role in identifying evidence and identifying deviations from the norm in various imaging modalities. The demand for this classification has increased along with the advancement of automation in all spheres of public life. OCT uses ultrasonic waves as a source and transmits the equivalent to produce images of the various retinal layer thicknesses [15].

A. Retinal Diseases

The advanced imaging method known as optical coherence tomography (OCT) uses logical light sources to provide images of optical diffusion medium with micrometer-level realism. The method is similar to ultrasonography. For OCT scans, light source are used rather than sound waves. Medical imaging offers many uses for optical coherence tomography. Early recognition and cure of eye disorders like - are being revolutionised by detailed images from retinal OCT scans. The retinal diseases include:

- Macular Hole (MH)
- Central Serous Retinopathy (CSR)
- Pigment Epithelial Detachment (PED)

1) Macular Hole

A macular hole is a tiny crack in the macula, which is part of the retina, the light-sensitive tissue of the eye. Our

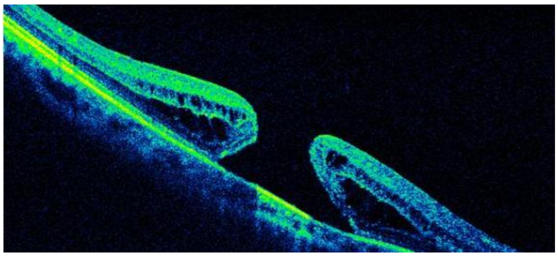


Fig. 1. Macular Hole

acute, centre vision required for reading, driving, and discerning minute details is provided by the macula. Central vision that is distorted and blurry can result from a macular hole.

Macular holes are an aging-related condition that often affects adults over 60. Despite having similar symptoms, age-related macular degeneration and macular holes are two different and distinct disorders. Both ailments are widespread in adults 60 and older. An expert in eye care will be able to tell the difference. The extent to which a hole will impair someone's vision depends on its size and position on the retina. The majority of central and fine detail vision may be lost when a Stage III macular hole develops. A detached retina, a disorder that threatens vision and needs emergency medical intervention, can result from an untreated macular hole.

2) Central Serous Retinopathy

A medical disorder called central serous retinopathy causes fluid to accumulate in the eye's retina. Because to the central retina's detachment, visual loss may occur suddenly or gradually. The macula is the name for this central region. A buildup of fluid behind the retina inside the eye is known as central serous retinopathy.

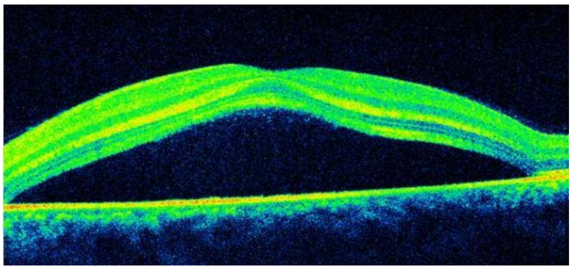


Fig. 2. Central Serous Retinopathy

The retina is in charge of turning the light that enters the eye into visuals that the brain can comprehend. Liquid accumulation can lead to the retina detaching, which can impair vision. Sometimes no treatment is necessary, and the patient will regain their vision within a short time. However, if someone starts to detect changes in their vision, they should visit their doctor right away. A typical sign is blurry vision. A person may experience the illness in both eyes at different times throughout their lifetime. There are times when central serous retinopathy goes unnoticed. It is possible for fluid to accumulate in locations other than the macula, which is in charge of providing sharp central vision. If this occurs, a person could not even be aware they have the ailment because they don't exhibit any symptoms.

3) Pigment Epithelial Detachment

Division between the RPE and the innermost part of Bruch's membrane is a characteristic of retinal pigment epithelial detachments (PEDs). Blood, serous exudate, drusenoid material, fibrovascular tissue, or a combination of these substances fill the gap left by this separation. Based on their composition, PEDs in AMD can be categorised into different groups. Drusenoid, serous, vascularized, or mixed components are some of the categories. Drusenoid PEDs are typically found in dry or nonneovascular AMD. Although serous PEDs are frequently linked to the wet or neovascular form of AMD, their natural history is generally more favourable. The risk of vision loss is higher in vascularized PEDs connected to Type 1 (sub-RPE) neovascularization and wet AMD. Multiple PED types are frequently observed in eyes with AMD [2].

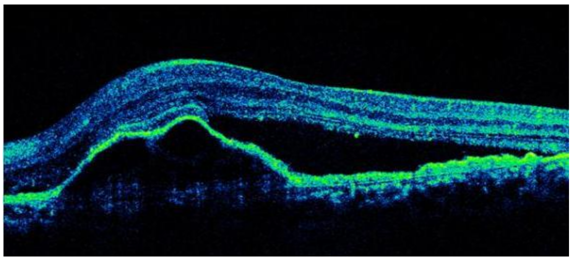


Fig. 3. Pigment Epithelial Detachment

Long-term PED has been linked to the emergence of choroidal neovascularization (CNV). Secondary to its elevated risk for serious visual loss, recognition of its presence is a major problem [3]. The Bruch's membrane's inner side has a structural split that allows the retinal pigment epithelium (RPE) to be actually separated from the remaining Bruch's membrane.

B. Artificial Neural Network

The term "artificial neural network" is derived from the biological neural networks that define the architecture of the human brain. Similar to how neurons are connected to one another in the real brain, artificial neural networks similarly include interconnected neurons at various layers of the networks. These neurons are referred to as nodes. Artificial Neural Network primarily consists of three layers.

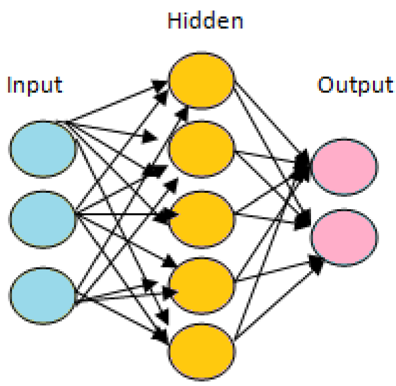


Fig. 4. ANN Architecture

1) Input Layer

As the name suggests, it accepts inputs in a number of formats that the programmer supplies.

2) Hidden Layer

The hidden layer lies between the input and output layers. It does all the calculations required to find hidden patterns and features.

3) Output Layer

The input is changed by the hidden layer into a number of different outputs, which are subsequently sent through this layer. The artificial neural network computes the weighted sum of the inputs and includes a bias when provided input. This algorithm is visualised using a transfer function.

$$\sum_{i=1}^{n} Wi * Xi + b \tag{1}$$

It feeds the weighted total as an input to an activation function, which outputs the result. The activation functions of a node control whether it should fire or not. Only those who are fired have access to the output layer. There are numerous activation functions that can be used, depending on the sort of task we are accomplishing [4].

II. Literature Review

In this work [19], a routine recognition and categorization system for retinal diseases has been developed to support doctors in their identification. A deep multilayered convolutional neural network (CNN) has been used to recognise and organize the retinal abnormality using OCT scans. [20] We look on the possibility of using patient retinal OCT scan images to automatically recognise and classify retinal abnormalities. We develop an approach to locate the region-of-interest in an OCT image of the retina, and we use a computationally light single layer convolutional neutral network structure for categorization. [21]. This work aims to discover and evaluate such anomaly in the retinal layers by using a classifier to recognise the anomaly that may serve as a prelude to loss of sight. Around 114 images from the OCT Modality were evaluated using MATLAB's SVM Classifier for common characteristics. After the SVM classifier was developed and evaluated to classify the input photographs as normal and Abnormal, the sensitivity and specificity were computed to evaluate the system's performance.

III. Proposed Method

Fig.5 depicts the block diagram of the suggested methodology. The region-of-interests (ROIs) from the retinal OCT images have first been segmented. Then the segmented ROIs have been use as input to the Neural Network. The retinal OCT scans are then divided into four classifications by the NN, including MH, CSR, PED, and NORMAL.

A. Preprocessing

Preprocessing is applied to the input photos, which includes noise removal and grayscale conversion. OCT uses ultrasound as a source of imaging, hence the images are more likely to have multiplicative speckle noises[16]–[18]. Weiner filters were employed to reduce/eliminate these disturbances by filtering out the speckle noises. Weiner filters are preferred in preprocessing over other filters despite the multiplicative character of the detected noises, specifically speckles.

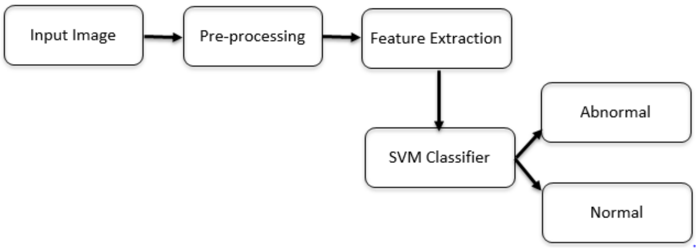
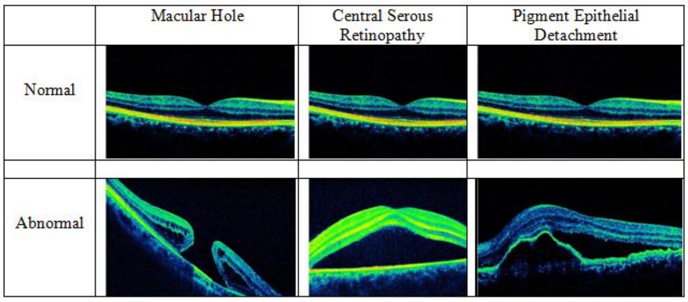


Fig. 5: Block Diagram of Proposed System



(a) (b) (c)

Fig 6. Normal and Abnormal Eye Conditions, a) Macular Hole, b) Central serous Retinopathy, c) Pigment Epithelial Detachment.

B. Segmentation

As shown in Fig. 7, there is a distinct Region of Interest where emphasis could be placed to find retinal anomalies for each type of disease. The Region-of-Interest in the OCT images was separated out using some fundamental image processing techniques. The pre-processing (segmentation) of the photos will help to improve data processing speed, decrease data loss, and increase accuracy.

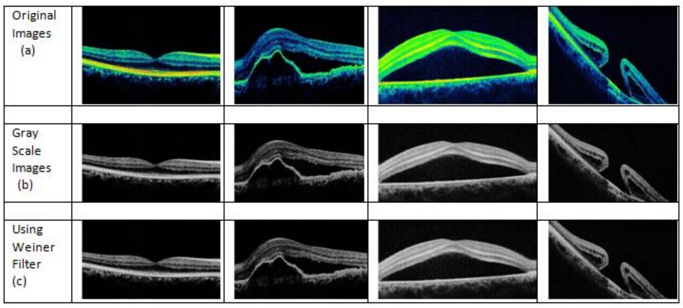


Fig. 7: (a) Original images, (b) Gray Scale and (c) Weiner filtering images

The segmenting procedure made use of the Python package OpenCV [9]. The dataset's photos had white borders around a sizable portion of them. These white borders would obstruct segmentation since a Binary Thresholding procedure is performed to identify the Region- of- Interest in the scan. As a result, dark pixels were used in place of the bright ones in the bordering areas. We utilised inverted Otsus Binarization to threshold the image [11]. As seen in Fig. 8, thresholding created a binary image with white pixels outside of the retinal regions and dark pixels inside of them. The noise in the image was decreased using a gap operation, followed by a dilation, but the thresholding procedure left some noise behind. Both procedures were carried out with a 33 unit kernel. The best results were obtained after 2 opening iterations and 7 dilation iterations. We were left with a clear binary image after the morphological alteration processes, as shown in Fig. 9.

Fig. 8. Binary Images

Fig. 9. Morphological close operation on binary image

IV. Results

Precision, recall, and f1-score, which are validating classification metrics, were used to further validate the accuracy metrics that the neural network training had produced. Table I tabulates accuracy metrics on the test set for several classes. A confusion matrix on the predicted labels against the actual labels for the various disease classifications can be created to further explain the findings. Fig. 10 shows corresponding accuracy metrics.

TABLE I

ACCURACY METRICS

	Precision	Recall	F1 - Score
Normal	0.97	0.99	0.98
MH	0.92	0.98	0.96
CSR	0.98	0.89	0.97
PED	0.98	0.98	0.97

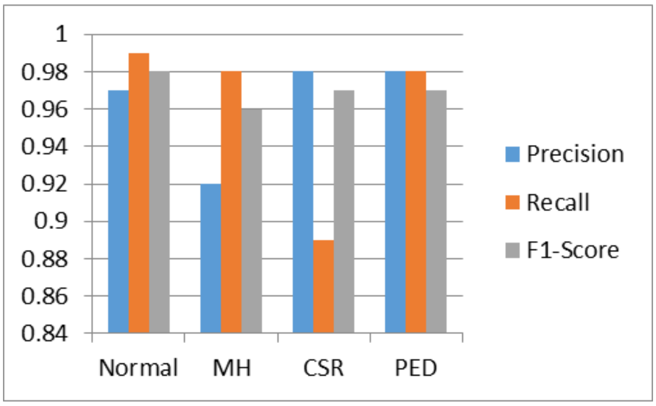


Fig. 10. Accuracy metrics

The confusion matrix for various disease classifications is shown in Table II without normalisation. Fig. 11 shows the comparison of accuracy.

TABLE II

CONFUSION MATRIX

		Predicted Label			
True Labels		Normal	MH	CSR	PED
	Normal	100	0	0	5
	MH	0	50	0	1
	CSR	9	20	50	2
	PED	0	5	0	50

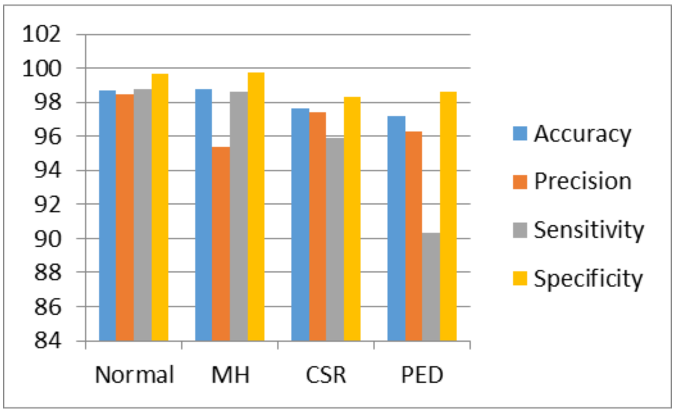


Fig. 11. Comparison of accuracy

CSR has received the most inaccurate labels out of the 150 images in that class, which has resulted in a small recall score. However, the outcomes from the other courses are highly compelling. The class by class accuracy metrics of our classifier were compared to those of Kermany et al. [10], who used transfer learning on the same dataset without any pre-processing, using a 48-layer deep computationally complex Inception-v3 architecture [13]. On the pre-processed image, our network, in contrast, just contains a single convolution and pooling layer, followed by two dense layers of neurons.

V. CONCLUSION

The proposed system and classification algorithms appear to be effective for classifying data in binary format. The degree of precision attained demonstrates that the technology can also be used for practical purposes. For specific abnormality-based classifications like Macular Hole, Central Serious Retinopathy, and Pigment Epithelial Detachment, the effectiveness of the proposed classifier systems needs to be assessed. Future age-related macular degeneration research could properly and thoroughly analyse the real performance. The created system can also be connected with the OCT devices in real time to automatically remark on the images that are provided as the input to the system, according to a detailed disorder-based categorization study.

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